Model Selection for OpenClaw Agent & Subagent Roles — 32GB VRAM Tier

OpenClaw Agent Framework — Model Analysis (Part 4) Date: February 14, 2026

Executive Summary

This report maps specific LLM models to OpenClaw agent and subagent roles within the 32GB VRAM tier. Every model is evaluated on its architecture (dense vs MoE), quantization options, context window capacity, speed characteristics under multi-GPU inference, and suitability for each agent role.

Key findings:

- **All 27–36B models from Part 3 are deployable** in OpenClaw agent roles at the 32GB tier — including Qwen3-32B, DeepSeek-R1-Distill-32B, Gemma 3 27B, Seed-OSS-36B, and EXAONE 4.0 32B.
- **Pipeline parallelism (layer split) is the method** for running these models across 2× 16GB GPUs when they exceed single-card VRAM. Every model except GPT-oss-20B requires this split.
- **MoE models are the optimal architecture for split deployments** — Qwen3-30B-A3B retains 92% of single-card speed when split, vs only 30% for dense models. This makes MoE the default choice for latency-sensitive agent roles.
- **llama.cpp and Ollama are the recommended inference tools** for multi-GPU pipeline splits. vLLM for production serving, ExLlamaV3 for maximum quantization quality.
- **Qwen3-30B-A3B is the recommended primary workhorse** for most OpenClaw roles, with dense models reserved for specialized tasks (deep reasoning, creative writing).
- **Dual-model serving** enables simultaneous operation of a fast router (8B) alongside a specialist model (14–20B) — eliminating model swap latency in multi-agent pipelines.

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1. Available Models — Architecture & Specifications

1.1 Complete Model Roster for 32GB Tier

Model	Architecture	Total Params	Active Params	Recommended Quant	Weight Size	Max Context on 32GB
Qwen3-32B	Dense	32B	32B	Q4_K_M / Q5_K_M	19.8 / 23.2 GB	16–45K tokens

Model	Architecture	Total Params	Active Params	Recommended Quant	Weight Size	Max Context on 32GB
DeepSeek-R1-Distill-32B	Dense	32B	32B	Q4_K_M	~20 GB	8–16K tokens
Gemma 3 27B	Dense	27B	27B	QAT Q4_0 / Q6_K / Q8_0	15.6 / 22 / 29 GB	20–40K+ (Q4) / 4–8K (Q8)
Qwen3-30B-A3B	MoE	30.5B	3.3B	Q4_K_M	~18.6 GB	32–131K tokens
Qwen3-Coder-30B-A3.3B	MoE	30.5B	3.3B	Q4_K_M	~18.6 GB	32–131K tokens
EXAONE 4.0 32B	Dense	32B	32B	Q4_K_M / Q5_K_M	19.8 / 23 GB	12–24K tokens
Seed-OSS-36B	Dense	36B	36B	Q4_K_M / Q5_K_M	21.5 / 25.6 GB	8–16K tokens
Nemotron 3 Nano 30B	Hybrid MoE	30B	~8B	UD-Q4_K_XL	~22.8 GB	16–32K tokens
GPT-oss-20B	MoE	21B	~4B	MXFP4 / Q4_K_M	~13.7 GB	Full context (fits single 16GB)

1.2 Why Architecture Matters for Agent Deployment

The choice between dense and MoE models has massive implications for OpenClaw agent performance in multi-GPU configurations:

- **MoE models (Qwen3-30B-A3B, Qwen3-Coder-30B, GPT-oss-20B, Nemotron):** Only a fraction of parameters activate per token (e.g., 3.3B of 30.5B for Qwen3-30B-A3B). This sparse activation dramatically reduces inter-GPU data transfer during inference. Result: **92% speed retention** when split across GPUs at short context.
- **Dense models (Qwen3-32B, R1-Distill-32B, Gemma 3 27B, EXAONE, Seed-OSS):** All parameters activate every token. Full weight data must flow through inter-GPU links. Result: **30% speed retention** when split — functional but significantly slower.

Practical takeaway: Default to MoE models for any latency-sensitive OpenClaw role. Reserve dense models for roles where their specific capabilities (deep reasoning, multimodal vision, creative quality) justify the speed tradeoff.

1.3 Higher Quantization Options at 32GB

The 32GB tier enables higher-quality quantizations that don't fit at 16GB, directly improving model output quality:

Model	Best Quant at 16GB	Best Quant at 32GB	Quality Gain
Qwen3-32B	❌ Doesn't fit	Q4_K_M or Q5_K_M	Enables model entirely; Q5_K_M adds ~2% coding quality
Gemma 3 27B	Q4_K_M (tight, minimal context)	Q6_K or Q8_0 (near-lossless)	+3–8% quality across benchmarks
DeepSeek-R1-Distill-32B	❌ Doesn't fit	Q4_K_M	Enables model entirely
Qwen3-30B-A3B	Q4_K_M (fits, but no KV room)	Q4_K_M + large KV cache	Context jumps from ~8K to 131K

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2. Pipeline Parallelism — How Models Exceeding 16GB Get Split

2.1 The Problem: 27–36B Models Don't Fit a Single 16GB Card

Every model in section 1 (except GPT-oss-20B) has weights exceeding 16GB at recommended quantization. Qwen3-32B Q4_K_M = 19.8 GB. Gemma 3 27B Q6_K = 22 GB. Seed-OSS-36B Q4_K_M = 21.5 GB. None of these load onto a single 16GB GPU. Pipeline parallelism solves this by distributing model layers across two GPUs.

2.2 How Pipeline Parallelism (Layer Split) Works

A model with 64 transformer layers gets split sequentially: layers 0–31 load on GPU 0, layers 32–63 load on GPU 1. During inference:

1. GPU 0 processes its layers, produces activation data (~8–16 KB)
2. Activation data transfers to GPU 1 through the PCIe bus
3. GPU 1 processes its layers, produces the output token
4. For the next token, the cycle repeats

Key characteristic: Only one GPU computes at a time — while GPU 1 works, GPU 0 sits idle (the "pipeline bubble"). This means you gain **VRAM capacity** (fit larger models), but **not speed** (token generation ≈ single-GPU speed). The value proposition is fitting 27–36B models that are otherwise impossible at 16GB.

2.3 Why MoE Models Suffer Less From Splitting

Dense models activate all parameters every token — the full 32B weights flow through computation. MoE models like Qwen3-30B-A3B only activate 3.3B of 30.5B parameters per token. The inactive experts sit in VRAM but don't participate in the forward pass, dramatically reducing the data that needs to cross the GPU boundary.

This is why the speed difference is so extreme:

- **Qwen3-30B-A3B (MoE):** ~101 t/s on split GPUs vs ~110 t/s on single 32GB card = **92% retained**
- **Qwen3-32B (dense):** ~18 t/s on split GPUs vs ~61 t/s on single 32GB card = **30% retained**

2.4 Alternative: Tensor Parallelism

Instead of splitting layers sequentially, tensor parallelism splits each layer's weight matrices across both GPUs so they compute simultaneously. In theory this is faster, but it requires heavy synchronization after every layer. Over consumer PCIe links (~32 GB/s effective), the synchronization overhead usually makes tensor parallelism **slower** than pipeline parallelism for token generation.

Exception: ExLlamaV3's tensor parallelism implementation is well-optimized and achieves 86% of single-RTX-3090 speed for MoE models even over PCIe. Worth testing if using EXL2 quantization.

2.5 Per-Model Split Behavior

Model	Weight Size	Fits Single 16GB?	Split Required?	Split Speed Impact
Qwen3-32B Q4_K_M	19.8 GB	❌ No	Yes — pipeline split	~30% of single-card speed (dense)
Qwen3-32B Q5_K_M	23.2 GB	❌ No	Yes — pipeline split	~28% of single-card speed (dense)
DeepSeek-R1-Distill-32B Q4_K_M	~20 GB	❌ No	Yes — pipeline split	~30% est. (dense)
Gemma 3 27B QAT Q4_0	15.6 GB	⚠️ Barely (no KV room)	Recommended	~35% est. (dense, smaller model)
Gemma 3 27B Q6_K	22 GB	❌ No	Yes — pipeline split	~30% est. (dense)
Gemma 3 27B Q8_0	29 GB	❌ No	Yes — pipeline split	~25% est. (dense, heavy weights)
Qwen3-30B-A3B Q4_K_M	18.6 GB	❌ No	Yes — pipeline split	**~92% of single-card speed (MoE)**

Model	Weight Size	Fits Single 16GB?	Split Required?	Split Speed Impact
Qwen3-Coder-30B Q4_K_M	18.6 GB	❌ No	Yes — pipeline split	**~92% est. (MoE)**
EXAONE 4.0 32B Q4_K_M	19.8 GB	❌ No	Yes — pipeline split	~30% est. (dense)
Seed-OSS-36B Q4_K_M	21.5 GB	❌ No	Yes — pipeline split	~25% est. (dense, largest model)
Nemotron 3 Nano 30B UD-Q4_K_XL	22.8 GB	❌ No	Yes — pipeline split	~60% est. (hybrid MoE)
GPT-oss-20B MXFP4	13.7 GB	✅ Yes	**No — keep on single GPU**	Splitting would reduce speed unnecessarily

3. Inference Tools for Multi-GPU Model Splitting

3.1 Which Tools Support Pipeline Parallelism for Split Models?

Not every inference engine handles multi-GPU splitting. Below is the compatibility matrix focused on what matters for OpenClaw deployment: can it split a 27–36B model across two 16GB GPUs, and how well does it work?

Tool	Split Method	Quantization Formats	Auto-Split?	Recommended For
llama.cpp	Pipeline (`-sm layer`) + Row (`-sm row`) + Graph (`-sm graph` in ik_llama.cpp fork)	GGUF (Q4_K_M, Q5_K_M, Q6_K, Q8_0)	Manual (`--tensor-split`)	Default choice — widest model support, mature multi-GPU
Ollama	Pipeline (auto, via llama.cpp backend)	GGUF (auto-downloaded from registry)	✅ Fully automatic	Simplest path — auto-detects GPUs, auto-distributes layers
vLLM	Pipeline (`--pipeline-parallel-size 2`) or Tensor (`--tensor-parallel-size 2`)	Safetensors, AWQ, GPTQ, FP8	Manual config	Production serving — OpenAI-compatible API, batched inference
ExLlamaV2/V3	Tensor (`--gpu_split auto`)	EXL2 (adaptive quantization)	✅ Auto GPU split	Best quant quality per bit — EXL2 outperforms GGUF at same bpw
SGLang	Pipeline, Tensor, Expert Parallelism, Data Parallelism	Safetensors, AWQ, GPTQ, FP8	Manual config	MoE-optimized expert parallelism — overkill for most setups
TensorRT-LLM	Tensor + Pipeline	FP8, INT8, INT4 (compiled engines)	No — requires engine compilation	Maximum speed — but hard setup, compile per model

3.2 Tool Recommendations by OpenClaw Use Case

Starting out / quick testing → Ollama Zero configuration. Install Ollama, run `ollama run qwen3:30b-a3b`, it auto-detects both GPUs and splits the model. Downside: only pipeline parallelism, no fine-tuning of split ratios.

Daily OpenClaw deployment → llama.cpp server Best balance of control and simplicity. Pipeline split with `-sm layer` is the recommended mode for consumer PCIe. Supports explicit VRAM allocation per GPU via `--tensor-split`. Runs as OpenAI-compatible API server for agent integration. Widest GGUF model selection.

Production / multi-user serving → vLLM Use `--pipeline-parallel-size 2` (not tensor parallel) for consumer GPUs — pipeline has lower communication overhead on PCIe. Continuous batching, OpenAI-compatible API. Best when OpenClaw runs as a service handling concurrent requests.

Maximum model quality → ExLlamaV3 + TabbyAPI EXL2 quantization delivers superior quality per bit vs GGUF. Tensor parallelism via `--gpu_split auto`. Benchmark: Qwen3-30B-A3B 5-bit on dual GPUs achieved

86% of single-RTX-3090 speed (44 t/s vs 51 t/s at 32K context). Use when output quality matters more than ecosystem convenience.

MoE-heavy workloads → SGLang Expert parallelism distributes MoE experts across GPUs more efficiently than simple layer split. If running Qwen3-30B-A3B or Qwen3-Coder-30B as the primary agent and want maximum MoE throughput, SGLang's expert parallelism is worth the setup complexity.

3.3 Tool Selection per Model

Model	Best Tool	Why	Alternative
Qwen3-30B-A3B Q4_K_M	llama.cpp / Ollama	GGUF ecosystem, pipeline split works great for MoE	SGLang (expert parallelism)
Qwen3-Coder-30B Q4_K_M	llama.cpp / Ollama	Same MoE architecture, same tool recommendation	vLLM (production serving)
Qwen3-32B Q4_K_M/Q5_K_M	llama.cpp	Pipeline split, GGUF — dense model, accept speed penalty	ExLlamaV3 (EXL2 for quality)
DeepSeek-R1-Distill-32B Q4_K_M	llama.cpp	Pipeline split, reasoning model — speed not critical	vLLM (if serving as API)
Gemma 3 27B Q6_K/Q8_0	llama.cpp / Ollama	GGUF, pipeline split, straightforward	ExLlamaV3 (EXL2 quality edge)
EXAONE 4.0 32B Q4_K_M	llama.cpp	GGUF pipeline split	vLLM (safetensors)
Seed-OSS-36B Q4_K_M	llama.cpp	Largest dense model at tier, needs reliable split	—
Nemotron 3 Nano 30B	llama.cpp	Hybrid MoE, GGUF support	SGLang (expert parallelism)
GPT-oss-20B MXFP4	llama.cpp (single GPU)	Fits one card — do NOT split	Ollama (single GPU)

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4. Complete Agent Role Mapping — All 5 Tiers (40+ Roles)

All speeds below are for models split via pipeline parallelism across 2× 16GB GPUs. For models that fit a single 16GB card (8B and below), speeds reflect single-GPU operation. Roles marked ⬆ **UPGRADE** benefit meaningfully from the 32GB tier vs 16GB single-card. Roles marked ➡ **SAME** run identically to 16GB — the 8B/14B model already handles them well and doesn't need splitting.

4.1 Tier 1 — Simple Roles (8 roles)

Simple classification, routing, or template-based tasks. Small models (0.6B–14B) handle these on a single 16GB card — no pipeline split needed. These roles do NOT benefit from the 32GB tier.

Role	Recommended Model	Quant	Speed	Split?	Status
Router/triage	Qwen3-8B	Q4_K_M	~186 t/s	No — single GPU	➡ SAME
Input validator	Qwen3-8B	Q4_K_M	~186 t/s	No — single GPU	➡ SAME
Heartbeat/health	Qwen3-0.6B	Q8_0	~500+ t/s	No — single GPU	➡ SAME
Notification/alert	Qwen3-8B	Q4_K_M	~186 t/s	No — single GPU	➡ SAME
Sentiment analysis	Qwen3-8B	Q4_K_M	~186 t/s	No — single GPU	➡ SAME
FAQ generation	Qwen3-14B	Q4_K_M	~124 t/s	No — single GPU	➡ SAME
Translation	Qwen3-14B	Q4_K_M	~124 t/s	No — single GPU	➡ SAME
Calendar/scheduling	Qwen3-8B	Q4_K_M	~186 t/s	No — single GPU	➡ SAME

Tier 1 summary: All 8 roles stay on a single 16GB card. No pipeline split needed. Second GPU is free for a second simultaneous model.

4.2 Tier 2 — Moderate Roles (25 roles)

Tasks requiring better comprehension, longer context, or more nuanced generation. Several roles upgrade at 32GB due to access to 27–36B models via pipeline split, especially document-heavy and analytical tasks.

Role	Recommended Model	Quant	Speed	Split?	Status	Notes
Research/web search	Qwen3-30B-A3B	Q4_K_M	~101 t/s	Yes — pipeline split	⬆ UPGRADE	MoE speed + 131K context for multi-source synthesis
Content writing	Qwen3-32B	Q5_K_M	~18 t/s	Yes — pipeline split	⬆ UPGRADE	Better prose quality at Q5. Dense = slow but worth it for writing
Editing/proofreading	Qwen3-32B	Q5_K_M	~18 t/s	Yes — pipeline split	⬆ UPGRADE	More consistent style corrections at higher quant
Email drafting	Qwen3-14B	Q6_K	~100 t/s	No — single GPU	➡ SAME	14B sufficient for email
Document summarization	Gemma 3 27B	Q6_K	~20 t/s	Yes — pipeline split	⬆ UPGRADE	Q6_K quality + 20–40K context handles longer docs
Meeting notes	Qwen3-30B-A3B	Q4_K_M	~101 t/s	Yes — pipeline split	⬆ UPGRADE	131K context for long meeting transcripts
Social media	Qwen3-14B	Q6_K	~100 t/s	No — single GPU	➡ SAME	Short-form content, 14B sufficient
News aggregation	Qwen3-14B	Q6_K	~100 t/s	No — single GPU	➡ SAME	Classification + summary, 14B handles well
Shopping/price comparison	Qwen3-8B	Q4_K_M	~186 t/s	No — single GPU	➡ SAME	Structured data extraction, small model fine
Memory/knowledge mgmt	Qwen3-30B-A3B	Q4_K_M	~101 t/s	Yes — pipeline split	⬆ UPGRADE	131K context + better retrieval accuracy

Role	Recommended Model	Quant	Speed	Split?	Status	Notes
RAG/retrieval	Qwen3-30B-A3B	Q4_K_M	~101 t/s	Yes — pipeline split	⬆ UPGRADE	Long context for retrieved chunks, higher extraction precision
Data analysis	Qwen3-32B	Q4_K_M	~18 t/s	Yes — pipeline split	⬆ UPGRADE	Better numerical reasoning at 32B
Website scraping	Qwen3-8B	Q4_K_M	~186 t/s	No — single GPU	➡ SAME	HTML parsing, structured extraction
Image description	Gemma 3 27B	Q8_0	~15 t/s	Yes — pipeline split	⬆ UPGRADE	Near-lossless multimodal — major quality jump vs Q4 on 16GB
Customer support	Qwen3-14B	Q6_K	~100 t/s	No — single GPU	➡ SAME	Conversation management, 14B sufficient
Lead scoring	Qwen3-14B	Q4_K_M	~124 t/s	No — single GPU	➡ SAME	Classification task
Project summarization	Qwen3-30B-A3B	Q4_K_M	~101 t/s	Yes — pipeline split	⬆ UPGRADE	131K context for full project docs
Transaction/approval	Qwen3-8B	Q4_K_M	~186 t/s	No — single GPU	➡ SAME	Simple validation/routing
Home automation	Qwen3-8B	Q4_K_M	~186 t/s	No — single GPU	➡ SAME	Command parsing, structured output
Fitness tracking	Qwen3-8B	Q4_K_M	~186 t/s	No — single GPU	➡ SAME	Data logging + simple analysis

Role	Recommended Model	Quant	Speed	Split?	Status	Notes
Recipe/cooking	Qwen3-14B	Q4_K_M	~124 t/s	No — single GPU	➡ SAME	Knowledge retrieval + generation
Personal finance	Qwen3-14B	Q6_K	~100 t/s	No — single GPU	➡ SAME	Calculation + advice, 14B handles well
SEO optimization	Qwen3-14B	Q6_K	~100 t/s	No — single GPU	➡ SAME	Content analysis + suggestions
Landing page generation	Qwen3-32B	Q5_K_M	~18 t/s	Yes — pipeline split	⬆ UPGRADE	Better code + copy quality at 32B
Travel planning	Qwen3-14B	Q6_K	~100 t/s	No — single GPU	➡ SAME	Itinerary generation, 14B sufficient

Tier 2 summary: 12 of 25 roles upgrade at 32GB. The upgrades are concentrated in document-heavy, analytical, and content creation roles where 27–36B models via pipeline split provide meaningfully better output. 13 roles remain on single-GPU 8B/14B models.

4.3 Tier 3 — Complex Roles (11 roles)

Tasks requiring strong coding, analytical reasoning, or multi-step synthesis. Almost all roles benefit from 27–36B models at 32GB. These are the roles that most justify the pipeline split.

Role	Recommended Model	Quant	Speed	Split?	Status	Notes
Code generation	Qwen3-Coder-30B-A3.3B	Q4_K_M	~100 t/s	Yes — pipeline split	⬆ UPGRADE	Purpose-built MoE coder. 131K context for large codebases
Code review	Qwen3-Coder-30B-A3.3B	Q4_K_M	~100 t/s	Yes — pipeline split	⬆ UPGRADE	Better at spotting subtle bugs at 30B vs 14B
QA/test writing	Qwen3-Coder-30B-A3.3B	Q4_K_M	~100 t/s	Yes — pipeline split	⬆ UPGRADE	More consistent test coverage generation
Task planning	Qwen3-30B-A3B	Q4_K_M	~101 t/s	Yes — pipeline split	⬆ UPGRADE	MoE speed for rapid planning iterations
Fact-checking	Qwen3-32B	Q5_K_M	~18 t/s	Yes — pipeline split	⬆ UPGRADE	Reduced hallucination at higher quant
Critic/review	Qwen3-32B	Q5_K_M	~18 t/s	Yes — pipeline split	⬆ UPGRADE	More nuanced analytical output
Market research	Qwen3-30B-A3B	Q4_K_M	~101 t/s	Yes — pipeline split	⬆ UPGRADE	131K context for multi-source synthesis
Synthesizer/aggregator	Qwen3-30B-A3B	Q4_K_M	~101 t/s	Yes — pipeline split	⬆ UPGRADE	Handles more input documents in single pass
Curriculum design	Qwen3-32B	Q5_K_M	~18 t/s	Yes — pipeline split	⬆ UPGRADE	Better coherence in structured educational content
Prototype generation	Qwen3-Coder-30B-A3.3B	Q4_K_M	~100 t/s	Yes — pipeline split	⬆ UPGRADE	Fast MoE coding for rapid prototyping

Role	Recommended Model	Quant	Speed	Split?	Status	Notes
DevOps	Qwen3-Coder-30B-A3.3B	Q4_K_M	~100 t/s	Yes — pipeline split	⬆ UPGRADE	More reliable YAML/config/IaC generation

Tier 3 summary: All 11 roles upgrade at 32GB. MoE models (Qwen3-Coder-30B, Qwen3-30B-A3B) handle 7 of 11 roles at ~100 t/s with pipeline split — excellent speed despite dual-GPU. Dense models (Qwen3-32B) serve 4 roles at ~18 t/s — slower but the quality gain justifies the tradeoff.

4.4 Tier 4 — Reasoning Roles (3 roles)

Deep multi-step reasoning, mathematical proofs, scientific analysis. Latency tolerance is high — quality trumps speed. Dense models are mandatory here because chain-of-thought reasoning benefits from full parameter activation.

Role	Recommended Model	Quant	Speed	Split?	Status	Notes
Math/logic reasoning	DeepSeek-R1-Distill-32B	Q4_K_M	~15–18 t/s	Yes — pipeline split	⬆ UPGRADE	R1-distilled chain-of-thought. Alt: EXAONE 4.0 32B (AIME 85.3%)
STEM analysis	DeepSeek-R1-Distill-32B	Q4_K_M	~15–18 t/s	Yes — pipeline split	⬆ UPGRADE	Better scientific reasoning fidelity at 32B
Algorithm exploration	Qwen3-32B (thinking mode)	Q4_K_M	~18 t/s	Yes — pipeline split	⬆ UPGRADE	Higher-quality code + reasoning combination

Tier 4 summary: All 3 roles upgrade at 32GB. Speed is slow (~15–18 t/s) because these are dense models on pipeline split, but reasoning chains are background tasks where quality matters far more than latency. These roles are impossible on a single 16GB card — the models simply don't fit.

4.5 Tier 5 — Frontier Roles (10 roles)

The hardest agent tasks: orchestrating complex multi-agent workflows, novel architecture decisions, legal/medical/financial analysis. These roles benefit from 32GB upgrades but cloud APIs (Claude, GPT-4.1, Gemini 2.5 Pro) still outperform local models for the most demanding instances.

Role	Recommended Model	Quant	Speed	Split?	Cloud still better?	Status
Orchestrator/manager	Qwen3-30B-A3B	Q4_K_M	~101 t/s	Yes — pipeline split	Yes, for complex orchestration	⬆ UPGRADE
Software architect	Qwen3-32B	Q5_K_M	~18 t/s	Yes — pipeline split	Yes, for novel architectures	⬆ UPGRADE
Complex debugger	Qwen3-Coder-30B-A3.3B	Q4_K_M	~100 t/s	Yes — pipeline split	Yes, for deep debugging	⬆ UPGRADE
Legal document review	Qwen3-32B	Q5_K_M	~18 t/s	Yes — pipeline split	Yes, strongly	⬆ UPGRADE
Medical/health analysis	Qwen3-32B	Q5_K_M	~18 t/s	Yes — pipeline split	Yes, always verify	⬆ UPGRADE
Financial analysis	Qwen3-32B	Q5_K_M	~18 t/s	Yes — pipeline split	Yes, for complex analysis	⬆ UPGRADE
Security analyst	Qwen3-Coder-30B-A3.3B	Q4_K_M	~100 t/s	Yes — pipeline split	Yes, for novel threats	⬆ UPGRADE
SRE/incident response	Qwen3-Coder-30B-A3.3B	Q4_K_M	~100 t/s	Yes — pipeline split	Partially	⬆ UPGRADE
Book writing	Qwen3-32B	Q5_K_M	~18 t/s	Yes — pipeline split	Yes, for style/coherence	⬆ UPGRADE
Compliance/regulatory	Qwen3-32B	Q5_K_M	~18 t/s	Yes — pipeline split	Yes, strongly	⬆ UPGRADE

Tier 5 summary: All 10 roles upgrade at 32GB. MoE models cover 4 roles (orchestrator, debugger, security, SRE) at ~100 t/s. Dense Qwen3-32B covers 6 roles at ~18 t/s — the quality gain from Q5_K_M is essential for these precision-critical tasks. No role transitions from "requires cloud API" to "fully solved locally" — the upgrade is incremental, not transformational.

4.6 Complete Summary — All Roles by Tier

Tier	Total Roles	Upgrade at 32GB	Stay Same	Need Pipeline Split	Best Architecture
Tier 1 (simple)	8	0	8	0	Single GPU (8B/14B)
Tier 2 (moderate)	25	12	13	12	Mixed: single GPU + split
Tier 3 (complex)	11	11	0	11	Pipeline split (MoE preferred)
Tier 4 (reasoning)	3	3	0	3	Pipeline split (dense required)
Tier 5 (frontier)	10	10	0	10	Pipeline split (mixed MoE + dense)
Total	**57**	**36**	**21**	**36**	—

Key takeaway: 36 of 57 roles (63%) benefit from the 32GB pipeline-split tier. All Tier 3–5 roles upgrade. The remaining 21 roles (all Tier 1 and half of Tier 2) run fine on single-GPU 8B/14B models and don't need splitting at all.

4.7 Model Usage Across All Roles

Model	Roles Served	Tier Coverage	Why It's Used
Qwen3-30B-A3B Q4_K_M (MoE)	12 roles	T2–T5	Primary workhorse. 101 t/s MoE speed + 131K context. Orchestrator, research, planning, RAG, memory, meetings, market research, synthesis
Qwen3-Coder-30B-A3.3B Q4_K_M (MoE)	8 roles	T3–T5	Coding specialist. ~100 t/s MoE speed. Code gen, review, QA, prototyping, DevOps, debugging, security, SRE
Qwen3-32B Q5_K_M (dense)	12 roles	T2–T5	Quality-critical tasks. ~18 t/s (slow). Writing, editing, fact-checking, critic, legal, medical, financial, compliance, book writing, architecture, landing pages

Model	Roles Served	Tier Coverage	Why It's Used
DeepSeek-R1-Distill-32B Q4_K_M (dense)	2 roles	T4	Reasoning specialist. ~15–18 t/s. Math/logic, STEM analysis
Gemma 3 27B Q6_K/Q8_0 (dense)	3 roles	T2	Multimodal + docs. ~15–20 t/s. Image description (Q8_0), document summarization (Q6_K), project summary
Qwen3-14B Q4_K_M/Q6_K	12 roles	T1–T2	Mid-range single GPU. ~100–124 t/s. Email, social media, customer support, finance, SEO, travel, recipes, news
Qwen3-8B Q4_K_M	8 roles	T1–T2	Fast single GPU. ~186 t/s. Router, validator, notifications, sentiment, scheduling, scraping, home automation, shopping
Qwen3-0.6B Q8_0	1 role	T1	Heartbeat/health monitor. ~500+ t/s. Ultra-lightweight
GPT-oss-20B MXFP4 (MoE)	Backup for multiple	T2–T3	Fits single 16GB card. Alternative workhorse when pipeline split not desired

5. Deployment Architectures for Multi-Agent Pipelines

5.1 Architecture A — Single Large Model (Split Across GPUs)

Both GPUs cooperate to run one 27–36B model via pipeline parallelism. Swap models between tasks.

Best for: Workloads dominated by a single role (e.g., heavy coding sessions using Qwen3-Coder-30B all day).

Model loading strategy:

1. Load Qwen3-30B-A3B as default workhorse (covers orchestrator, research, planning)
2. Swap to Qwen3-Coder-30B for coding sessions
3. Swap to R1-Distill-32B for deep reasoning tasks
4. Swap to Gemma 3 27B Q8_0 for vision tasks

Swap time: ~5–15 seconds per model load depending on storage speed (NVMe recommended).

5.2 Architecture B — Dual Independent Models (One Per GPU)

Each GPU runs a separate model simultaneously. Zero swap latency between roles.

Example configurations:

GPU 0 (16GB)	GPU 1 (16GB)	Use Case
Qwen3-8B Q4_K_M (~5.5 GB)	GPT-oss-20B MXFP4 (~13.7 GB)	Fast router + specialist workhorse
Qwen3-14B Q4_K_M (~9 GB)	Gemma 3 27B QAT Q4_0 (~15.6 GB)	General assistant + vision/multilingual
Qwen3-Coder-8B Q4_K_M (~5.5 GB)	Qwen3-14B Q5_K_M (~10.5 GB)	Code specialist + general reasoning
Qwen3-0.6B draft (~0.5 GB)	Qwen3-14B Q4_K_M (~9 GB)	Speculative decoding (1.5–2× throughput)

Best for: Multi-agent pipelines where different subagents need different models and model swap latency is unacceptable. The orchestrator can route to whichever GPU hosts the appropriate model.

5.3 Architecture C — Hybrid (Split + Small Sidecar)

Split a 20–25GB model across both GPUs, leaving room for a small sidecar model on one GPU.

Example: Qwen3-30B-A3B Q4_K_M (~18.6 GB split) + Qwen3-0.6B (~0.5 GB on GPU 0) for speculative decoding. The draft model proposes tokens, the split model verifies — potential 1.5–2× throughput improvement.

5.4 Recommended Architecture by Workload

Workload Pattern	Best Architecture	Why
Single-role focus (coding all day)	A — Single split model	Maximum model quality, no switching overhead
Multi-agent pipeline (OpenClaw full stack)	B — Dual independent	Zero swap latency, parallel subagent serving
High-throughput single model	C — Hybrid with speculative decoding	1.5–2× throughput boost
Mixed workloads with vision	A — Swap between Qwen3-30B-A3B and Gemma 3 27B	Different models for different tasks

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6. Model Performance by Agent Role — Detailed Analysis

6.1 Speed Tiers for Multi-GPU Deployment

Models fall into distinct speed tiers that determine their suitability for interactive vs background agent roles:

Tier 1 — Fast Interactive (~80–110 t/s on multi-GPU):

- Qwen3-30B-A3B Q4_K_M: ~101 t/s
- Qwen3-Coder-30B-A3.3B Q4_K_M: ~100 t/s (est.)
- Nemotron 3 Nano 30B: ~60 t/s (est.)

These models are suitable for **interactive agent roles** where the user is waiting: orchestrator, code generation, planning, real-time chat.

Tier 2 — Moderate (~20–35 t/s on multi-GPU):

- Gemma 3 27B QAT Q4_0: ~25 t/s (est.)
- Gemma 3 27B Q6_K: ~20 t/s (est.)

Suitable for **semi-interactive roles** with moderate latency tolerance: document summary, multilingual processing, research analysis.

Tier 3 — Slow (~15–18 t/s on multi-GPU):

- Qwen3-32B Q4_K_M/Q5_K_M: ~18 t/s
- DeepSeek-R1-Distill-32B Q4_K_M: ~15–18 t/s (est.)
- EXAONE 4.0 32B Q4_K_M: ~18 t/s (est.)
- Seed-OSS-36B Q4_K_M: ~15 t/s (est.)
- Gemma 3 27B Q8_0: ~15 t/s (est.)

Suitable for **background agent roles** where quality trumps speed: deep reasoning, creative writing, vision tasks, specialized analysis.

6.2 Context Window Analysis by Role

Context length determines how much information an agent can process in a single pass. This is critical for OpenClaw roles handling large inputs.

Agent Role	Context Needed	Best Model	Available Context
Research (full documents)	32K–128K	Qwen3-30B-A3B Q4_K_M	**131K** ✅
Code Generation (large repos)	16K–64K	Qwen3-Coder-30B Q4_K_M	**131K** ✅
Orchestrator (task state)	8K–32K	Qwen3-30B-A3B Q4_K_M	**131K** ✅
Document Summary	16K–48K	Gemma 3 27B Q6_K	**20–40K** ✅
Deep Reasoning	4K–16K	R1-Distill-32B Q4_K_M	**8–16K** ✅
Creative Writing	4K–16K	Qwen3-32B Q5_K_M	**16–32K** ✅
Vision (image + text)	2K–8K	Gemma 3 27B Q8_0	**4–8K** ✅
Planning	8K–32K	Qwen3-30B-A3B Q4_K_M	**131K** ✅

Key observation: Qwen3-30B-A3B's 131K context at MoE speed is a standout advantage. No dense model in this tier comes close to that context-to-speed ratio.

6.3 Quality Benchmarks by Domain

Relative model quality rankings for each agent domain (based on public benchmark data and community evaluations):

Coding (HumanEval, MBPP, LiveCodeBench):

1. Qwen3-Coder-30B-A3.3B — purpose-built, top coding MoE
2. Qwen3-32B Q5_K_M — strong all-rounder, benefits from Q5 quant
3. DeepSeek-R1-Distill-32B — good at complex algorithmic reasoning
4. Seed-OSS-36B — competitive, benefits from larger parameter count

Reasoning (MATH, GPQA, ARC-Challenge):

1. DeepSeek-R1-Distill-32B — distilled from R1, chain-of-thought specialist
2. Qwen3-32B — strong general reasoning
3. Seed-OSS-36B — competitive at 36B scale
4. Qwen3-30B-A3B — good but MoE routing adds slight overhead for deep chains

General Language (MMLU, HellaSwag, WinoGrande):

1. Qwen3-32B — best general language at this tier

2. EXAONE 4.0 32B — competitive, strong multilingual
3. Gemma 3 27B — excellent comprehension
4. Seed-OSS-36B — broad capability

Vision/Multimodal (MMMU, DocVQA, ChartQA):

1. Gemma 3 27B Q8_0 — near-lossless vision, best quality
2. Gemma 3 27B Q6_K — very close to Q8, more context room
3. Gemma 3 27B QAT Q4_0 — acceptable for basic visual tasks

Multilingual (Flores, multilingual MMLU):

1. Gemma 3 27B — 100+ languages
2. EXAONE 4.0 32B — excellent CJK languages
3. Qwen3-32B — strong Chinese/English bilingual

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7. Models That Don't Fit the 32GB Tier

For reference, these models from adjacent tiers cannot be deployed at 32GB:

Model	Total Size (Q4_K_M)	Minimum VRAM	Notes
Llama 4 Scout (109B MoE)	~65 GB	4× 16GB (64GB)	MoE but too many total params
Llama 3.3 70B (dense)	~40 GB	48GB+	Needs 2× 24GB or single 48GB
Qwen3-235B-A22B (MoE)	~85.7 GB (Q2_K)	6+ GPUs	Datacenter scale
GPT-oss-120B (MoE)	~59 GB	4× 16GB (64GB)	Large MoE
DeepSeek-R1 (671B MoE)	~140 GB+	Datacenter	Multi-node deployment

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8. Conclusion & Recommended OpenClaw Model Stack

8.1 Primary Recommendation

The optimal model stack for OpenClaw at the 32GB tier covers all 57 agent roles across 5 tiers. MoE models handle speed-sensitive roles; dense models serve quality-critical tasks:

Priority	Model	Roles Covered	Key Advantage
Primary workhorse	Qwen3-30B-A3B Q4_K_M	12 roles (T2–T5): orchestrator, research, planning, RAG, memory, meetings, market research, synthesis	101 t/s + 131K context
Coding specialist	Qwen3-Coder-30B-A3.3B Q4_K_M	8 roles (T3–T5): code gen, review, QA, prototyping, DevOps, debugging, security, SRE	Purpose-built MoE coder, ~100 t/s
Quality writer	Qwen3-32B Q5_K_M	12 roles (T2–T5): writing, editing, fact-checking, legal, medical, financial, compliance, architecture, book writing	Best language quality at tier
Deep reasoning	DeepSeek-R1-Distill-32B Q4_K_M	2 roles (T4): math/logic, STEM analysis	R1-distilled chain-of-thought
Multimodal + docs	Gemma 3 27B Q6_K/Q8_0	3 roles (T2): image description, document summary, project summary	Near-lossless vision at Q8_0
Mid-range general	Qwen3-14B Q4_K_M/Q6_K	12 roles (T1–T2): email, social media, customer support, finance, SEO, travel	Fast single-GPU, no split needed
Fast lightweight	Qwen3-8B Q4_K_M	8 roles (T1): router, validator, notifications, sentiment, scheduling, scraping	186 t/s, single GPU

8.2 Core Insight: MoE-First Strategy

The single most important principle for OpenClaw model selection at 32GB in multi-GPU configurations:

- **Qwen3-30B-A3B (MoE):** 92% speed retention when split across GPUs
- **Qwen3-32B (dense):** 30% speed retention when split across GPUs

This 3× difference in speed retention means the optimal strategy is **Qwen3-30B-A3B as primary workhorse** for most agent roles, reserving dense models only for roles where their specific capabilities are irreplaceable.

8.3 Minimum Viable Stack

For the simplest effective OpenClaw deployment at 32GB:

1. **Qwen3-30B-A3B Q4_K_M** — covers orchestrator, planning, research, general tasks, and serviceable coding (20+ roles across T2–T5)
2. **Gemma 3 27B Q6_K** — covers vision, multilingual, and document processing (3+ roles)
3. **Qwen3-8B Q4_K_M** on single GPU — handles all Tier 1 simple roles simultaneously (8 roles)

4. Swap to Qwen3-32B Q5_K_M or R1-Distill-32B when quality-critical writing or deep reasoning is needed

This three-model rotation covers all 57 roles. Load Qwen3-30B-A3B as the default split model and keep Qwen3-8B on one GPU as a fast router — the MoE workhorse handles the majority of tasks at 101 t/s.